microduck-banana-pcb-locker SCALE 2:1 A1 THIRD ANGLE

ISOMETRIC 1 ? FRONT-LEFT ? 1000 x 324 px, RENDERED OFF THIS SOLID

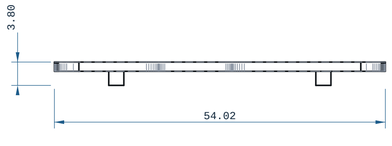
ISOMETRIC 2 ? FRONT-RIGHT ? 1000 x 513 px, RENDERED OFF THIS SOLID

ISOMETRIC 3 ? REAR-RIGHT ? 1000 x 513 px, RENDERED OFF THIS SOLID

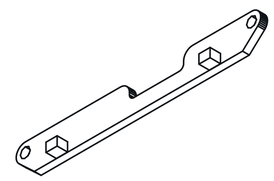
ISOMETRIC 4 ? REAR-LEFT ? 1000 x 394 px, RENDERED OFF THIS SOLID

TOP-DOWN RENDER ? 1000 x 394 px, RENDERED OFF THIS SOLID

BOTTOM-UP RENDER ? 1000 x 394 px, RENDERED OFF THIS SOLID

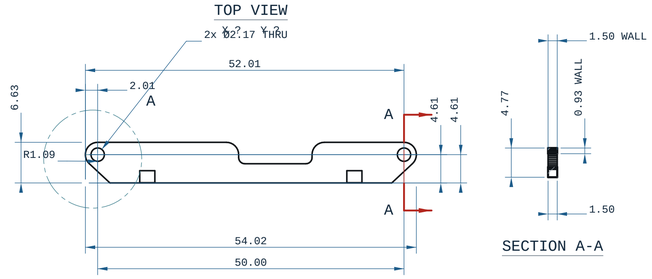


TOP VIEW



ISOMETRIC VIEW

54.02 x 3.80 x 6.63 mm



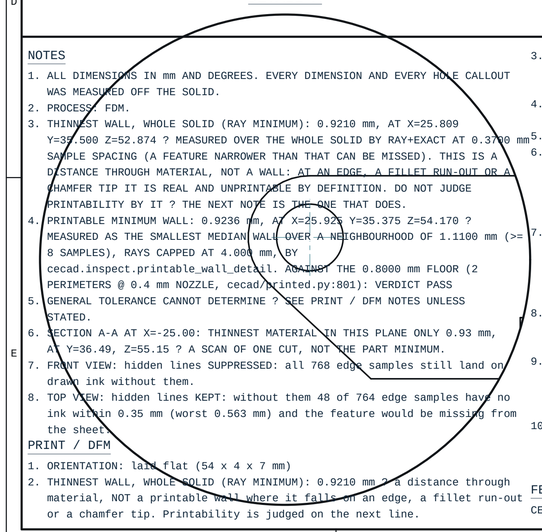
FRONT VIEW

-X ? Z ?

SECTION A-A

SCALE 10:1

DETAIL A



NOTES

1. ALL DIMENSIONS IN mm AND DEGREES. EVERY DIMENSION AND EVERY HOLE CALLOUT WAS MEASURED OFF THE SOLID.
2. PROCESS: FDM.
3. THINNEST WALL, WHOLE SOLID (RAY MINIMUM): 0.9210 mm, AT X=25.809 Y=36.500 Z=52.874 ? MEASURED OVER THE WHOLE SOLID BY RAY+EXACT AT 0.3700 mm SAMPLE SPACING (A FEATURE NARROWER THAN THAT CAN BE MISSED). THIS IS A DISTANCE THROUGH MATERIAL, NOT A WALL: AT AN EDGE, A FILLET RUN-OUT OR A CHAMFER TIP IT IS REAL AND UNPRINTABLE BY DEFINITION. DO NOT JUDGE PRINTABILITY BY IT ? THE NEXT NOTE IS THE ONE THAT DOES.
4. PRINTABLE MINIMUM WALL: 0.9236 mm, AT X=25.925 Y=35.375 Z=54.170 ? MEASURED AS THE SMALLEST MEDIAN WALL OVER A NEIGHBOURHOOD OF 1.1100 mm (>= 8 SAMPLES), RAYS CAPPED AT 4.000 mm, BY cecad.inspect.printable_wall_detail. AGAINST THE 0.8000 mm FLOOR (2 PERIMETERS @ 0.4 mm NOZZLE, cecad/printed.py:801): VERDICT PASS
5. GENERAL TOLERANCE CANNOT DETERMINE ? SEE PRINT / DFM NOTES UNLESS STATED.
6. SECTION A-A AT X=-25.00: THINNEST MATERIAL IN THIS PLANE ONLY 0.93 mm, AT Y=36.49, Z=55.15 ? A SCAN OF ONE CUT, NOT THE PART MINIMUM.
7. FRONT VIEW: hidden lines SUPPRESSED: all 768 edge samples still land on drawn ink without them.
8. TOP VIEW: hidden lines KEPT: without them 48 of 764 edge samples have no ink within 0.35 mm (worst 0.563 mm) and the feature would be missing from the sheet.

PRINT / DFM

1. ORIENTATION: laid flat (54 x 4 x 7 mm)
2. THINNEST WALL, WHOLE SOLID (RAY MINIMUM): 0.9210 mm ? a distance through material, NOT a printable wall where it falls on an edge, a fillet run-out or a chamfer tip. Printability is judged on the next line.
3. PRINTABLE MINIMUM WALL: 0.9236 mm (smallest median wall over a 1.1100 mm neighbourhood, cecad.inspect.printable_wall_detail) against a 0.8000 mm floor (2 perimeters @ 0.4 mm nozzle): VERDICT PASS
4. 2 horizontal bore(s) bridge at the crown: ream any bearing or press-fit bore to size
5. FILAMENT: PLA, ~1 g (modelled), 34 layers @ 0.2 mm
6. TOLERANCE BASIS, IN-HOUSE: CANNOT DETERMINE. Every Bambu on this farm records tolerance_mm: null, cite "no coupon printed and measured on this machine" (ce-machines/bambu-h2s-*, bambu-h2d-*/capabilities.json, read 2026-09-02), and Bambu Lab's own H2D Pro spec sheet carries no dimensional-accuracy line. DIMENSIONS HERE ARE NOMINAL AS MODELLED.
7. TOLERANCE BASIS, OUTSOURCED: JLC3DP ±0.3 mm FDM (jlc3dp.com, ce-machines/farm-jlc3dp); Hubs ±0.5%, floor ±0.5 mm (hubs.com/3d-printing/, ce-machines/farm-hubs). Apply the route's figure, not both. The ±0.2 mm in cecad/machining.py fdm is a repo planning constant with no vendor cite and is NOT used here. One printed and measured coupon closes this note.
8. NO ISO 286 CLASS IS DECLARED ON ANY BORE: FDM cannot be sized to one (H7/p6 at Ø3 is a 0.010 mm band). REAM bearing and press-fit bores after printing.
9. RADII NAMED IN cad/part.py THAT ARE NOT CIRCULAR EDGES ON THIS SOLID: 1.00 mm, 2.00 mm. cecad.inspect.arc_radii measures 1 ARC(S) HERE AND EVERY ONE IS DIMENSIONED ABOVE. NO DIMENSION IS MISSING: A LEADER NEEDS AN EDGE. THE GEOMETRY FILE CARRIES THE SURFACE.
10. ORTHOGRAPHIC VIEWS: 2 (FRONT, TOP) ? cecad.autosheet.choose_views ON THE 54.015 x 3.800 x 6.626 mm BBOX, THIN/LONG 0.0704. A THIRD VIEW WOULD BE THIS PART EDGE-ON, WHICH A.2 ORDERS SUPPRESSED; THE SECTION AND THE ISOMETRIC CARRY IT.

FEATURE SCHEDULE ? 23 FEATURE(S) ENUMERATED OFF THE SOLID, GENERAL TOLERANCE CANNOT DETERMINE ? SEE PRINT / DFM NOTES

CENSUS cecad.sheetcheck.enumerate_features v2 ? 23 ROW(S), 6 CANNOT DETERMINE, LISTED BELOW

OUTLINE ENVELOPE LENGTH	54.0153
OUTLINE ENVELOPE WIDTH	3.8000
OUTLINE ENVELOPE HEIGHT	6.6260
HOLE 1 DIAMETER (ALONG Y)	Ø2.1740
HOLE 1 DEPTH	1.5000
HOLE 1 POSITION FROM THE X DATUM FACE	2.0076
HOLE 1 POSITION FROM THE Z DATUM FACE	4.6130
HOLE 2 DIAMETER (ALONG Y)	Ø2.1740
HOLE 2 DEPTH	1.5000
HOLE 2 POSITION FROM THE X DATUM FACE	52.0076
HOLE 2 POSITION FROM THE Z DATUM FACE	4.6130
RADIUS R1.09 (4 EDGES, 4 FULL CIRCLES)	R1.0900
THINNEST WALL	0.9210
POCKET MOUTH ON A PLANE NORMAL TO Y, 4 EDGE(S) ? WIDTH	1.9660
POCKET MOUTH ON A PLANE NORMAL TO Y, 4 EDGE(S) ? LENGTH	2.4660
POCKET MOUTH ON A PLANE NORMAL TO Y, 4 EDGE(S) ? DEPTH	2.3000
POCKET MOUTH ON A PLANE NORMAL TO Y, 4 EDGE(S) ? POSITIO	10.1076
POCKET MOUTH ON A PLANE NORMAL TO Y, 4 EDGE(S) ? POSITIO	1.0130
POCKET MOUTH ON A PLANE NORMAL TO Y, 4 EDGE(S) ? WIDTH	1.9660
POCKET MOUTH ON A PLANE NORMAL TO Y, 4 EDGE(S) ? LENGTH	2.4660
POCKET MOUTH ON A PLANE NORMAL TO Y, 4 EDGE(S) ? DEPTH	2.3000
POCKET MOUTH ON A PLANE NORMAL TO Y, 4 EDGE(S) ? POSITIO	43.9076
POCKET MOUTH ON A PLANE NORMAL TO Y, 4 EDGE(S) ? POSITIO	1.0130

CANNOT DETERMINE ? FEATURES THIS CENSUS COULD NOT MEASURE (A4)

CANNOT DETERMINE ? CHAMFER: A4 requires size and angle on every chamfer; cecad.inspect has no chamfer classifier ? a chamfer is a planar face bounding two faces at an angle and nothing here identifies one. SETTLED BY: a chamfer classifier in cecad.inspect (planar face whose two neighbours meet it at equal angles), or the builder declaring its chamfers in the part record

CANNOT DETERMINE ? RIB: A4 requires thickness, height and pitch on every rib; nothing here groups faces into a rib. SETTLED BY: a rib classifier (parallel thin prisms standing off a wall), or the builder declaring its rib pitch

CANNOT DETERMINE ? DRAFT: A4 requires every draft angle; no face here is tested for draft against a pull direction. SETTLED BY: a pull direction on the part record, then the angle of every planar face to it

CANNOT DETERMINE ? STEP: A4 requires every step and shoulder; nothing here classifies a shoulder between two coaxial diameters or two parallel faces. SETTLED BY: a step classifier, or the section views declaring their stations

CANNOT DETERMINE ? MATE: A4 requires every surface that mates with another part, with the mating part named by its triad ref; the mates live in ce-assemblies/*/joints.json and are not read here. SETTLED BY: reading this part's joints out of the assemblies that use it and matching each mate to a face

CANNOT DETERMINE ? WALL: A4 wants a wall thickness at every distinct station; only the thinnest wall is measurable off the solid without a station list. SETTLED BY: the drawer declaring the section stations it dimensions, or a section per shell region

MICRODUCK-BANANA-PCB-LOCKER

REV A

MATERIAL PLA		PROCESS FDM	SCALE 2:1
CATEGORY bracket	GENERAL TOL CANNOT DETER–	FINISH AS PRINTED	SHEET A1 1/1
VOLUME (MEASURED) 0.46 cm3		MASS (ASSUMES PLA) 0.6 g	UNITS mm / deg
SIZE (BBOX, MEASURED) 54.02 x 3.80 x 6.63		DRAWN ce-cad 2026-09-05	PROJECTION THIRD ANGLE
SOURCE (DESIGN FILE) ce-parts/microduck-banana-pcb-locker/current/cad/part.py			